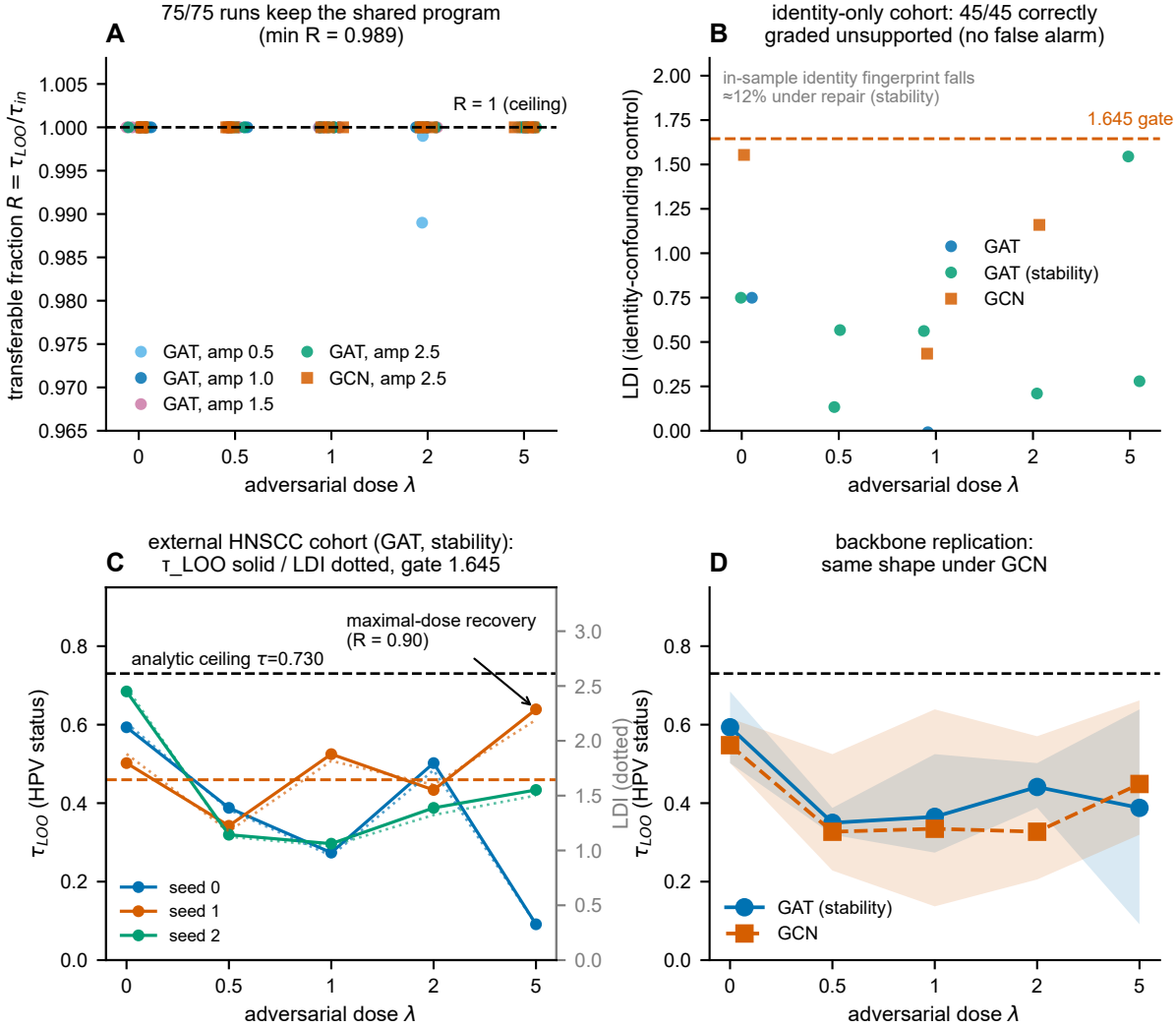

A
75/75 runs keep the shared program
(min R = 0.989)
transferable fraction $R = \tau_{LOO}/\tau_{in}$
R = 1 (ceiling)
GAT, amp 0.5
GAT, amp 1.0
GAT, amp 1.5
GAT, amp 2.5
GCN, amp 2.5
adversarial dose λ
B
identity-only cohort: 45/45 correctly
graded unsupported (no false alarm)
in-sample identity fingerprint falls
≈12% under repair (stability)
1.645 gate
LDI (identity-confounding control)
GAT
GAT (stability)
GCN
adversarial dose λ
C
external HNSCC cohort (GAT, stability):
τ_LOO solid / LDI dotted, gate 1.645
maximal-dose recovery
(R = 0.90)
analytic ceiling τ=0.730
τ_{LOO} (HPV status)
LDI (dotted)
seed 0
seed 1
seed 2
adversarial dose λ
D
backbone replication:
same shape under GCN
τ_{LOO} (HPV status)
GAT (stability)
GCN
adversarial dose λ